# Electrodynamics

# 电动力学

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Typeset with LaTeX

# 目录

# 1 电磁场的基本方程

**Maxwell 方程组**

$$
\begin{cases}
\nabla\cdot\boldsymbol{D}=\rho\\
\nabla\times\boldsymbol{E}=-\dfrac{\partial\boldsymbol{B}}{\partial t}\\
\nabla\cdot\boldsymbol{B}=0\\
\nabla\times\boldsymbol{H}=\boldsymbol{j}+\dfrac{\partial\boldsymbol{D}}{\partial t}
\end{cases}
$$

**本构方程**

$$
\begin{cases}
\boldsymbol{D}=\varepsilon\boldsymbol{E}\\
\boldsymbol{B}=\mu\boldsymbol{H}\\
\boldsymbol{j}=\sigma\boldsymbol{E}
\end{cases}
$$

**边界条件**

设 $\boldsymbol{n}$ 为介质 1 指向介质 2 的单位法向量

$$
\begin{cases}
\boldsymbol{n}\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1)=\sigma\\
\boldsymbol{n}\cdot(\boldsymbol{B}_2-\boldsymbol{B}_1)=0\\
\boldsymbol{n}\times(\boldsymbol{E}_2-\boldsymbol{E}_1)=0\\
\boldsymbol{n}\times(\boldsymbol{H}_2-\boldsymbol{H}_1)=\boldsymbol{\alpha}
\end{cases}
$$

# 2 静电场

**静电场求解思路**

$$
\begin{cases}
\nabla\cdot\boldsymbol{D}=\rho\\
\nabla\times\boldsymbol{E}=0
\end{cases}
\implies \boldsymbol{E}=-\nabla\varphi
$$

$$
\boldsymbol{E}(\boldsymbol{r})=\int\frac{\rho(\boldsymbol{x}')}{4\pi\varepsilon_0 r^2}\,\mathrm{d}V'
$$

$$
\varphi(\boldsymbol{r})=\int\frac{\rho(\boldsymbol{x}')}{4\pi\varepsilon_0 r}\,\mathrm{d}V'
$$

- 对于具有对称性的电场

$$\oint \boldsymbol{D}\cdot \mathrm{d}\boldsymbol{S} = Q \quad \Longrightarrow \boldsymbol{D} \quad \xRightarrow{\boldsymbol{D}=\varepsilon \boldsymbol{E}} \boldsymbol{E}$$

$$\varphi(r) = \int_r^{\infty} \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{l} = \int_r^{R_1} \boldsymbol{E}_1\cdot \mathrm{d}\boldsymbol{l} + \int_{R_1}^{R_2} \boldsymbol{E}_2\cdot \mathrm{d}\boldsymbol{l} + \int_{R_2}^{R_3} \boldsymbol{E}_3\cdot \mathrm{d}\boldsymbol{l} + \cdots$$

- 对于复杂问题, 一般需要求解 Poisson 方程

$$\nabla^2\varphi = -\frac{\rho}{\varepsilon}$$

- 当 $\rho = 0$ 时退化为 Laplace 方程

$$\nabla^2\varphi = 0$$

**分离变量法: 轴对称通解**

在球坐标下且具有轴对称性时, Laplace 方程 $\nabla^2\varphi = 0$ 通解的形式为

$$\varphi(r,\theta) = \sum_n \left(a_n r^n + \frac{b_n}{r^{n+1}}\right) \mathrm{P}_n(\cos\theta)$$

根据边界条件求出 $a_n, b_n$

**分离变量法: 边界条件**

- $r\to\infty$

$$\begin{cases} \varphi = \varphi_0 - E_0 r\cos\theta & \text{有外场} \\ \varphi = 0 & \text{无外场} \end{cases}$$

- $r\to 0$, $\varphi$ 有限

- $r\to R$ 的介质间分界面

$$\begin{cases} \varphi_1|_{r=R} = \varphi_2|_{r=R} \\ \varepsilon_1 \left.\dfrac{\partial\varphi_1}{\partial r}\right|_{r=R} = \varepsilon_2 \left.\dfrac{\partial\varphi_2}{\partial r}\right|_{r=R} \end{cases}$$

- $r\to R$ 的导体表面

$$\begin{cases} \varphi = \varphi_{\mathrm{s}} \\ \oint \sigma\,\mathrm{d}S = Q \quad \sigma = \boldsymbol{n}\cdot(\boldsymbol{D}_2 - \boldsymbol{D}_1) \end{cases}$$

## 镜像法

使用条件: $\begin{cases} \text{电荷数目有限} \\ \text{边界规则} \end{cases}$

镜像电荷的引入, 不能改变边界条件, 且镜像电荷不能在所求空间内. 镜像电荷的电荷量不一定等于边界上的感应电荷量, 但其效应完全等价于感应电荷产生的效应, 如电场、电势、作用力. 镜像电荷可以不止一个.

## 格林函数法

- 第一类边界条件 $\varphi = C|_S$

$$\varphi(\boldsymbol{x}) = \int_V \rho(\boldsymbol{x}')G(\boldsymbol{x}',\boldsymbol{x})\,\mathrm{d}V' - \varepsilon_0 \oint_S \varphi(\boldsymbol{x}')\frac{\partial G(\boldsymbol{x}',\boldsymbol{x})}{\partial n'}\,\mathrm{d}S'$$

- 第二类边界条件 $\left.\dfrac{\partial\varphi}{\partial n}\right|_S = C$

$$\varphi(\boldsymbol{x}) = \int_V \rho(\boldsymbol{x}')G(\boldsymbol{x}',\boldsymbol{x})\,\mathrm{d}V' + \varepsilon_0 \oint_S G(\boldsymbol{x}',\boldsymbol{x})\frac{\partial \varphi(\boldsymbol{x}')}{\partial n'}\,\mathrm{d}S' + \langle\varphi\rangle_S$$

- $G(\boldsymbol{x},\boldsymbol{x}')$ 为 Green 函数, 满足

$$\nabla^2 G(\boldsymbol{x},\boldsymbol{x}') = -\frac{1}{\varepsilon_0}\delta(\boldsymbol{x}-\boldsymbol{x}')$$

## 电多极展开

$$\varphi(\boldsymbol{x}) = \frac{Q}{4\pi\varepsilon_0 r} + \frac{\boldsymbol{p}\cdot\boldsymbol{r}}{4\pi\varepsilon_0 r^3} + \frac{1}{4\pi\varepsilon_0}\frac{1}{6}\mathcal{D}:\nabla\nabla\frac{1}{r} + \cdots$$

- 总电荷

$$Q = \int_V \rho(\boldsymbol{x}')\,\mathrm{d}V'$$

- 电偶极矩

$$\boldsymbol{p} = \int_V \rho(\boldsymbol{x}')\boldsymbol{x}'\,\mathrm{d}V'$$

- 电四极矩张量

$$\mathcal{D} = \int_V 3\boldsymbol{x}'\boldsymbol{x}'\rho(\boldsymbol{x}')\,\mathrm{d}V'$$

## 电偶极矩在外场中的能量

$$W = \boldsymbol{p}\cdot\nabla\varphi_{\mathrm{e}} = -\boldsymbol{p}\cdot\boldsymbol{E}_{\mathrm{e}}$$

**电四极矩在外场中的能量**

$$W = -\frac{1}{6}\mathcal{D} : \nabla \boldsymbol{E}_{\mathrm{e}}$$

# 3 静磁场

**静磁场基本方程**

$$\begin{cases} \nabla \times \boldsymbol{H} = \boldsymbol{J} \\ \nabla \cdot \boldsymbol{B} = 0 \end{cases}$$

**边值关系**

$$\begin{cases} \boldsymbol{n} \times (\boldsymbol{H}_2 - \boldsymbol{H}_1) = \boldsymbol{\alpha} \\ \boldsymbol{n} \cdot (\boldsymbol{B}_2 - \boldsymbol{B}_1) = 0 \end{cases}$$

**对称磁场的求解思路**

由 Ampère 环路定理

$$\oint \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{l} = I \quad \Rightarrow \boldsymbol{H} \quad \Rightarrow \boldsymbol{B} = \mu \boldsymbol{H}$$

**磁矢势**

由 $\nabla \cdot \boldsymbol{B} = 0$ 引入矢势

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} \qquad \boldsymbol{A} = \frac{\mu_0}{4\pi} \int_V \frac{\boldsymbol{J}(\boldsymbol{x}')}{r} \, \mathrm{d}V'$$

**磁标势**

当 $\boldsymbol{J} = 0$ 且 $\oint \boldsymbol{H} \cdot \mathrm{d}\boldsymbol{l} = 0$ (单连通区域) 时, 可引入磁标势

$$\boldsymbol{H} = -\nabla \varphi_{\mathrm{m}}$$

$$\nabla^2 \varphi_{\mathrm{m}} = -\frac{\rho_{\mathrm{m}}}{\mu_0} \qquad (\rho_{\mathrm{m}} = -\mu_0 \nabla \cdot \boldsymbol{M})$$

当 $\rho_{\mathrm{m}} = 0$ 时退化为 Laplace 方程

$$\nabla^2 \varphi_{\mathrm{m}} = 0$$

## 通解

与电势一样, 在球坐标下且具有轴对称性, 有通解

$$\varphi_{\mathrm{m}}(r,\theta)=\sum_n\left(a_nr^n+\frac{b_n}{r^{n+1}}\right)\mathrm{P}_n(\cos\theta)$$

## 边界条件

$$\begin{cases}\varphi_{\mathrm{m}1}\Big|_S=\varphi_{\mathrm{m}2}\Big|_S\\ \mu_1\dfrac{\partial\varphi_{\mathrm{m}1}}{\partial n}\Big|_S=\mu_2\dfrac{\partial\varphi_{\mathrm{m}2}}{\partial n}\Big|_S\end{cases}$$

$$\begin{cases}r\to 0 & \varphi_{\mathrm{m}}\ \text{有限}\\ r\to\infty & \begin{cases}\varphi_{\mathrm{m}}=-H_0r\cos\theta & \text{有外场}\\ \varphi_{\mathrm{m}}=0 & \text{无外场}\end{cases}\end{cases}$$

## 矢势矢势 $\boldsymbol{A}$ 的多极展开

$$\begin{aligned}\boldsymbol{A}(\boldsymbol{x})&=\frac{\mu_0}{4\pi}\int_V\frac{\boldsymbol{J}(\boldsymbol{x}')}{r}\,\mathrm{d}V'\\&=\frac{\mu_0}{4\pi}\int_V\boldsymbol{J}(\boldsymbol{x}')\left(\frac{1}{R}-(\boldsymbol{x}'\cdot\nabla)\frac{1}{R}+\frac{1}{2}(\boldsymbol{x}'\cdot\nabla)^2\frac{1}{R}+\cdots\right)\mathrm{d}V'\\&=\frac{\mu_0}{4\pi}\frac{\boldsymbol{m}\times\boldsymbol{R}}{R^3}+\cdots\end{aligned}$$

## 磁矩

$$\boldsymbol{m}=\frac{I}{2}\oint(\boldsymbol{x}'\times\mathrm{d}\boldsymbol{l})\qquad(\boldsymbol{m}=I\boldsymbol{S})$$

## AB 效应

在量子力学框架下, 矢势 $\boldsymbol{A}$ 具有可测量的物理意义. 即使在磁场为零的区域, 矢势仍可影响带电粒子的量子相位.

## 超导体

- 超导体的两大基本特性
  1. 零电阻;
  2. 完全抗磁性: Meissner 效应.
- 临界磁场, 临界电流, 磁通量子化.

- 临界磁场与温度的关系

$$H_{\mathrm{c}}(T) = H_{\mathrm{c}}(0)\left[1 - \left(\frac{T}{T_{\mathrm{c}}}\right)^2\right] \qquad (T \le T_{\mathrm{c}})$$

**London 方程**

**London 第一方程**

$$\frac{\partial \boldsymbol{J}_{\mathrm{s}}}{\partial t} = \alpha \boldsymbol{E} \qquad \alpha = \frac{n_{\mathrm{s}} e^2}{m}$$

**London 第二方程**

$$\nabla \times \boldsymbol{J}_{\mathrm{s}} = -\alpha \boldsymbol{B}$$

# 4 电磁场与物质相互作用

**极化强度**

$\boldsymbol{p}_i = q\boldsymbol{l}$ 为单个分子的电偶极矩.

$$\boldsymbol{P} = \frac{\sum \boldsymbol{p}_i}{\Delta V}$$

对各向同性均匀介质

$$\boldsymbol{P} = \chi_{\mathrm{e}} \varepsilon_0 \boldsymbol{E}$$

**极化电荷体密度**

$$\rho_{\mathrm{p}} = -\nabla \cdot \boldsymbol{P} \qquad Q_{\mathrm{p}} = -\oint \boldsymbol{P} \cdot \mathrm{d}\boldsymbol{S}$$

**极化面电荷密度**

$$\sigma_{\mathrm{p}} = -\boldsymbol{n} \cdot (\boldsymbol{P}_2 - \boldsymbol{P}_1)$$

**介质中的 Gauss 定理**

$$\oint \boldsymbol{D} \cdot \mathrm{d}\boldsymbol{S} = \int \rho \, \mathrm{d}V \qquad \nabla \cdot \boldsymbol{D} = \rho$$

**电位移矢量**

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P}$$

对各向同性均匀介质

$$\boldsymbol{D} = \varepsilon_0 \boldsymbol{E} + \boldsymbol{P} = \varepsilon_0 \boldsymbol{E} + \chi_{\mathrm{e}} \varepsilon_0 \boldsymbol{E} = \varepsilon_0 (1 + \chi_{\mathrm{e}}) \boldsymbol{E} = \varepsilon_0 \varepsilon_{\mathrm{r}} \boldsymbol{E} = \varepsilon \boldsymbol{E}$$

$\varepsilon_{\mathrm{r}}=1+\chi_{\mathrm{e}}$ 相对介电常数, $\varepsilon=\varepsilon_0\varepsilon_{\mathrm{r}}$ 介电常数.

## 极化电流

对交变电场, 极化电流密度

$$\boldsymbol{j}_{\mathrm{p}}=\frac{\partial \boldsymbol{P}}{\partial t}$$

## 磁化强度

$\boldsymbol{m}_i=i\boldsymbol{a}$ 为单个分子的磁矩

$$\boldsymbol{M}=\frac{\sum \boldsymbol{m}_i}{\Delta V}$$

对各向同性均匀介质

$$\boldsymbol{M}=\chi_{\mathrm{m}}\boldsymbol{H}$$

## 磁化电流密度

$$\boldsymbol{j}_{\mathrm{m}}=\nabla\times\boldsymbol{M}$$

## 磁化面电流密度

$$\boldsymbol{\alpha}_{\mathrm{m}}=\boldsymbol{n}\times(\boldsymbol{M}_2-\boldsymbol{M}_1)$$

## 介质中的 Ampère 环路定理

$$\oint \boldsymbol{H}\cdot\mathrm{d}\boldsymbol{l}=\int \boldsymbol{J}\cdot\mathrm{d}\boldsymbol{S}+\int\frac{\partial \boldsymbol{D}}{\partial t}\cdot\mathrm{d}\boldsymbol{S}$$

$$\nabla\times\boldsymbol{H}=\boldsymbol{J}+\frac{\partial \boldsymbol{D}}{\partial t}$$

## 磁场强度

$$\boldsymbol{H}=\frac{\boldsymbol{B}}{\mu_0}-\boldsymbol{M}=\frac{\boldsymbol{B}}{\mu}$$

## Coulomb 相互作用

$$\boldsymbol{F}=\frac{q_1q_2}{4\pi\varepsilon_0}\frac{\boldsymbol{r}}{r^3}$$

## 场对电荷的相互作用: Lorentz 力 (密度)

$$\boldsymbol{F}=q\boldsymbol{E}+q\boldsymbol{v}\times\boldsymbol{B}$$

$$\boldsymbol{f}=\rho\boldsymbol{E}+\boldsymbol{J}\times\boldsymbol{B}$$

# 5 电磁场的能量和动量

**能量密度**

$$w = \frac{1}{2}(\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{H}\cdot\boldsymbol{B})$$

$$\begin{cases} w = \dfrac{1}{2}\varepsilon E^2 + \dfrac{1}{2}\dfrac{B^2}{\mu} & \text{均匀介质} \\ w = \dfrac{1}{2}\varepsilon_0 E^2 + \dfrac{1}{2}\dfrac{B^2}{\mu_0} & \text{真空} \end{cases}$$

**能流密度 (Poynting 矢量)**

$$\boldsymbol{S} = \boldsymbol{E}\times\boldsymbol{H}$$

$$\langle \boldsymbol{S} \rangle = \frac{1}{2}\,\mathrm{Re}\{\boldsymbol{E}^*\times\boldsymbol{H}\}$$

**损耗功率**

$$\langle P \rangle = \frac{1}{2}\,\mathrm{Re}\{\boldsymbol{J}^*\cdot\boldsymbol{E}\}$$

**动量密度与动量流密度**

$$\boldsymbol{g} = \boldsymbol{D}\times\boldsymbol{B}$$

$$\mathcal{T} = -(\boldsymbol{DE} + \boldsymbol{BH}) + \frac{1}{2}\mathcal{I}\,(\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{B}\cdot\boldsymbol{H})$$

**能量守恒**

$$\frac{\partial w}{\partial t} + \boldsymbol{f}\cdot\boldsymbol{v} = -\nabla\cdot\boldsymbol{S}$$

**动量守恒**

$$\boldsymbol{f} + \frac{\partial}{\partial t}\boldsymbol{g} = -\nabla\cdot\mathcal{T}$$

**电荷守恒**

$$\frac{\partial \rho}{\partial t} + \nabla\cdot\boldsymbol{j} = 0$$

# 6 电磁波的传播

## 时谐电磁波

$$\boldsymbol{E}(\boldsymbol{x},t)=\boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}\qquad \boldsymbol{B}(\boldsymbol{x},t)=\boldsymbol{B}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}$$

## Helmholtz 方程

$$\begin{cases}\nabla\times\boldsymbol{E}=-\dfrac{\partial\boldsymbol{B}}{\partial t}\\ \nabla\times\boldsymbol{H}=\dfrac{\partial\boldsymbol{D}}{\partial t}\\ \nabla\cdot\boldsymbol{D}=0\\ \nabla\cdot\boldsymbol{B}=0\end{cases}\xrightarrow[k=\omega\sqrt{\mu\varepsilon}]{\partial_t\to-\mathrm{i}\omega}\begin{cases}\nabla^2\boldsymbol{E}+k^2\boldsymbol{E}=0\quad(\nabla\cdot\boldsymbol{E}=0)\\ \nabla^2\boldsymbol{B}+k^2\boldsymbol{B}=0\quad(\nabla\cdot\boldsymbol{B}=0)\end{cases}$$

- 求 $\boldsymbol{E}$

$$\begin{cases}\nabla^2\boldsymbol{E}+k^2\boldsymbol{E}=0\quad(\nabla\cdot\boldsymbol{E}=0)\\ \boldsymbol{B}=-\dfrac{\mathrm{i}}{\omega}\nabla\times\boldsymbol{E}\end{cases}$$

- 求 $\boldsymbol{B}$

$$\begin{cases}\nabla^2\boldsymbol{B}+k^2\boldsymbol{B}=0\quad(\nabla\cdot\boldsymbol{B}=0)\\ \boldsymbol{E}=\dfrac{\mathrm{i}}{\omega\mu\varepsilon}\nabla\times\boldsymbol{B}\end{cases}$$

## 平面电磁波

$$\boldsymbol{E}(\boldsymbol{x},t)=\boldsymbol{E}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}\qquad \boldsymbol{B}(\boldsymbol{x},t)=\boldsymbol{B}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}$$

1. 平面电磁波是 TEM 波, $\boldsymbol{k}\perp\boldsymbol{E}$, $\boldsymbol{k}\perp\boldsymbol{B}$;
2. $\boldsymbol{E}\perp\boldsymbol{B}$, $\boldsymbol{k}\parallel\boldsymbol{E}\times\boldsymbol{B}$, 三者满足右手螺旋关系;
3. $\boldsymbol{E}$ 与 $\boldsymbol{B}$ 同相位, 振幅比为相速度 $v$;
4. 电磁波传播速度 $v=1/\sqrt{\mu\varepsilon}$.

## 相速度与群速度

$$v_{\mathrm{p}}=\frac{\omega}{k}\qquad v_{\mathrm{g}}=\frac{\mathrm{d}\omega}{\mathrm{d}k}$$

## 圆偏振

$$\begin{cases} \boldsymbol{E} = E_0 \mathrm{e}^{\mathrm{i}(kz-\omega t)}(\boldsymbol{e}_x - \mathrm{i}\boldsymbol{e}_y) & \text{右旋} \\ \boldsymbol{E} = E_0 \mathrm{e}^{\mathrm{i}(kz-\omega t)}(\boldsymbol{e}_x + \mathrm{i}\boldsymbol{e}_y) & \text{左旋} \end{cases}$$

## 反射和折射定律

$$\theta = \theta' \qquad n_1 \sin\theta = n_2 \sin\theta''$$

## Fresnel 公式

$$\begin{cases} \dfrac{E'_\perp}{E_\perp} = -\dfrac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')} & \dfrac{E''_\perp}{E_\perp} = \dfrac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')} \\ \dfrac{E'_\parallel}{E_\parallel} = \dfrac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')} & \dfrac{E''_\parallel}{E_\parallel} = \dfrac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')\cos(\theta-\theta'')} \end{cases}$$

## 反射系数与透射系数

$$R = \left|\frac{E'}{E}\right|^2 \qquad T = 1 - R$$

## Brewster 角

$$\theta_\mathrm{B} + \theta''_\mathrm{B} = 90^\circ \quad \Longrightarrow \frac{E''_\parallel}{E_\parallel} = 0$$

$$\tan\theta_\mathrm{B} = \frac{n_2}{n_1}$$

## 全反射

$$\sin\theta > \sin\theta_\mathrm{c} = \frac{n_2}{n_1}$$

$$k''_z = \mathrm{i}\,\underbrace{k\sqrt{\sin^2\theta - (n_2/n_1)^2}}_{=\kappa},\ k''_x = k_x = k\sin\theta,\ \boldsymbol{E}'' = \boldsymbol{E}''_0 \mathrm{e}^{-\kappa z}\mathrm{e}^{\mathrm{i}(k''_x x - \omega t)}.$$

$$\kappa^{-1} = \frac{\lambda_1}{2\pi\sqrt{\sin^2\theta - (n_2/n_1)^2}} = \frac{\lambda}{2\pi n_1\sqrt{\sin^2\theta - (n_2/n_1)^2}}$$

$$v_\mathrm{p} = \frac{c}{n_1 \sin\theta}$$

## 良导体条件

$$\rho(t) = \rho_0 \mathrm{e}^{-\frac{\sigma}{\varepsilon}t}$$

$$\omega \ll \tau^{-1} = \frac{\sigma}{\varepsilon} \quad \Longrightarrow \frac{\sigma}{\varepsilon\omega} \gg 1$$

## 复介电常量与复波矢

$$\varepsilon' = \varepsilon + \mathrm{i}\frac{\sigma}{\omega}$$

$$k = \omega\sqrt{\mu\varepsilon'} \qquad \boldsymbol{k} = \boldsymbol{\beta} + \mathrm{i}\boldsymbol{\alpha}$$

## 趋肤深度

$$\delta = \frac{1}{\alpha} \approx \sqrt{\frac{2}{\omega\mu\sigma}}$$

## 理想导体壁边界条件

$$\begin{cases} \boldsymbol{n} \times \boldsymbol{E} = 0 \\ \dfrac{\partial E_n}{\partial n} = 0 \end{cases}$$

## 矩形谐振腔

$$\begin{cases} E_x = A_1 \cos k_x x \ \sin k_y y \ \sin k_z z \\ E_y = A_2 \sin k_x x \ \cos k_y y \ \sin k_z z \\ E_z = A_3 \sin k_x x \ \sin k_y y \ \cos k_z z \end{cases}$$

$$k_x = \frac{m\pi}{L_1}, \quad k_y = \frac{n\pi}{L_2}, \quad k_z = \frac{p\pi}{L_3} \qquad (m, n, p = 0, 1, 2, 3, \ldots)$$

由 $\nabla \cdot \boldsymbol{E} = 0$ 得

$$k_x A_1 + k_y A_2 + k_z A_3 = 0$$

一组 $(m, n, p)$ 对应两种独立的电磁波模式.

## 谐振频率 (本征频率)

$$\omega_{mnp} = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{L_1}\right)^2 + \left(\frac{n}{L_2}\right)^2 + \left(\frac{p}{L_3}\right)^2}$$

### 矩形波导

$$\begin{cases} E_x = A_1 \cos k_x x \ \sin k_y y \, \mathrm{e}^{\mathrm{i}(k_z z - \omega t)} \\ E_y = A_2 \sin k_x x \ \cos k_y y \, \mathrm{e}^{\mathrm{i}(k_z z - \omega t)} \\ E_z = A_3 \sin k_x x \ \sin k_y y \, \mathrm{e}^{\mathrm{i}(k_z z - \omega t)} \end{cases}$$

$$k_x = \frac{m\pi}{a}, \quad k_y = \frac{n\pi}{b} \qquad (m, n = 0, 1, 2, 3, \dots)$$

由 $\nabla \cdot \boldsymbol{E} = 0$ 得

$$k_x A_1 + k_y A_2 - \mathrm{i} k_z A_3 = 0$$

一组 $(m, n)$ 对应两种独立模式.
磁场模式:

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} = \mathrm{i}\omega\mu \boldsymbol{H}$$

波导中只能传播 TE 模或 TM 模 (不能传播 TEM 模).

### 截止频率与截止波长

$$\omega_{\text{cut}} = \frac{\pi}{\sqrt{\mu\varepsilon}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\lambda_{\text{cut}} = \frac{2ab}{\sqrt{(mb)^2 + (na)^2}}$$

### 等离子体

Debye 屏蔽 Coulomb 势

$$\varphi(r) = \frac{q}{4\pi\varepsilon_0 r} \mathrm{e}^{-r/\lambda_{\mathrm{D}}} \qquad \lambda_{\mathrm{D}} = \sqrt{\frac{\varepsilon_0 k T}{n e^2}}$$

等离子体的振荡频率

$$\omega_{\mathrm{p}} = \sqrt{\frac{n e^2}{m \varepsilon_0}}$$

等离子体中电磁波的传播

$$\frac{\partial \boldsymbol{j}}{\partial t} = \frac{n e^2}{m} \boldsymbol{E}$$

$$\boldsymbol{j}(\omega) = \sigma(\omega) \boldsymbol{E}(\omega) \qquad \sigma(\omega) = \mathrm{i} \frac{n e^2}{m \omega}$$

$\omega > \omega_{\mathrm{p}}$ 时电磁波才能在等离子体中传播, 故 $\omega_{\mathrm{p}}$ 即为截止频率.

# 7 电磁波辐射

**电磁场的矢势和标势**

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} \qquad \boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t} - \nabla \varphi$$

**常用的规范条件**

1. Coulomb 规范

$$\nabla \cdot \boldsymbol{A} = 0$$

2. Lorenz 规范

$$\nabla \cdot \boldsymbol{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t} = 0$$

**d'Alembert 方程**

$$\begin{cases} \nabla^2 \boldsymbol{A} - \dfrac{1}{c^2}\dfrac{\partial^2 \boldsymbol{A}}{\partial t^2} = -\mu_0 \boldsymbol{J} \\ \nabla^2 \varphi - \dfrac{1}{c^2}\dfrac{\partial^2 \varphi}{\partial t^2} = -\dfrac{\rho}{\varepsilon_0} \end{cases} \qquad \left(\nabla \cdot \boldsymbol{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t} = 0\right)$$

**推迟势**

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{\mu_0}{4\pi}\int_V \frac{\boldsymbol{J}(\boldsymbol{x}', t - r/c)}{r}\,\mathrm{d}V'$$

$$\varphi(\boldsymbol{x}, t) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\boldsymbol{x}', t - r/c)}{r}\,\mathrm{d}V'$$

**小区域辐射**

$$\text{条件: } l \ll \lambda,\ l \ll r. \begin{cases} \text{近区 (静态区): } r \ll \lambda, kr \ll 1 \\ \text{感应区 (过渡区): } r \sim \lambda, kr \sim 1 \\ \text{远区 (辐射区): } r \gg \lambda, kr \gg 1 \end{cases}$$

**远区辐射**

$$\begin{aligned} \boldsymbol{A}(\boldsymbol{x}) &= \frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi R}\int_V \boldsymbol{J}(\boldsymbol{x}')\left[1 - \mathrm{i}k\,\boldsymbol{n}\cdot\boldsymbol{x}' + \frac{1}{2!}(\mathrm{i}k\,\boldsymbol{n}\cdot\boldsymbol{x}')^2 + \cdots\right]\mathrm{d}V' \\ &= \boldsymbol{A}_{(1)}(\boldsymbol{x}) + \boldsymbol{A}_{(2)}(\boldsymbol{x}) + \cdots \end{aligned}$$

**电偶极辐射**

$$\boldsymbol{A}_{(1)}(\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\int_V\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}V'=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\dot{\boldsymbol{p}}$$

$$\boldsymbol{B}=\nabla\times\boldsymbol{A}=\frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi\varepsilon_0c^3R}\ddot{\boldsymbol{p}}\times\boldsymbol{e}_r=\frac{1}{4\pi\varepsilon_0c^3R}\mathrm{e}^{\mathrm{i}kR}|\ddot{\boldsymbol{p}}|\sin\theta\,\boldsymbol{e}_\varphi$$

$$\boldsymbol{E}=c\boldsymbol{B}\times\boldsymbol{e}_r=\frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi\varepsilon_0c^2R}(\ddot{\boldsymbol{p}}\times\boldsymbol{e}_r)\times\boldsymbol{e}_r=\frac{1}{4\pi\varepsilon_0c^2R}\mathrm{e}^{\mathrm{i}kR}|\ddot{\boldsymbol{p}}|\sin\theta\,\boldsymbol{e}_\theta$$

**电偶极辐射平均能流密度**

$$\boldsymbol{S}=\frac{|\ddot{\boldsymbol{p}}|^2}{32\pi^2\varepsilon_0c^3R^2}\sin^2\theta\,\boldsymbol{e}_r$$

**电偶极辐射全空间平均辐射总功率**

$$P=\frac{1}{4\pi\varepsilon_0}\frac{|\ddot{\boldsymbol{p}}|^2}{3c^3}$$

**短天线的辐射电阻**

$$R_\mathrm{r}=\frac{\pi}{6}\sqrt{\frac{\mu_0}{\varepsilon_0}}\left(\frac{l}{\lambda}\right)^2=197\left(\frac{l}{\lambda}\right)^2\Omega$$

**半波天线电流分布**

$$I(z)=\begin{cases}I_0\sin k\left(\dfrac{l}{2}-z\right) & 0\le z\le\dfrac{l}{2}\\[2ex] I_0\sin k\left(\dfrac{l}{2}+z\right) & -\dfrac{l}{2}\le z<0\end{cases}$$

**半波天线的矢势与辐射场**

$$A_z(\boldsymbol{x})=\frac{\mu_0I_0\mathrm{e}^{\mathrm{i}kR}}{2\pi kR}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$$

$$\boldsymbol{B}=\nabla\times\boldsymbol{A}=-\mathrm{i}\frac{\mu_0I_0\mathrm{e}^{\mathrm{i}kR}}{2\pi R}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin\theta}\boldsymbol{e}_\varphi$$

$$\boldsymbol{E}=c\boldsymbol{B}\times\boldsymbol{e}_r=-\mathrm{i}\frac{\mu_0cI_0\mathrm{e}^{\mathrm{i}kR}}{2\pi R}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin\theta}\boldsymbol{e}_\theta$$

**半波天线平均能流密度**

$$\boldsymbol{S}=\frac{1}{2}\mathrm{Re}(\boldsymbol{E}\times\boldsymbol{H}^*)=\frac{\mu_0cI_0^2}{8\pi^2R^2}\frac{\cos^2\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\boldsymbol{e}_r$$

**半波天线辐射总功率**

$$P = 2.44\,\frac{\mu_0 c I_0^2}{8\pi} = \frac{1}{2} I_0^2 R_{\mathrm{r}}$$

**半波天线辐射电阻**

$$R_{\mathrm{r}} = 2.44\,\frac{\mu_0 c}{4\pi} = 73.2\,\Omega$$

# 8 狭义相对论

**基本原理**

1. **相对性原理**:所有惯性参考系中物理定律的形式相同.
2. **光速不变原理**:真空中的光速在任何惯性系中均为恒定值 $c$, 与光源运动无关.

**Lorentz 变换**

设 $\Sigma'$ 系沿 $\Sigma$ 系的 $x$ 轴以速度 $v$ 运动

$$\begin{cases} x' = \dfrac{x - vt}{\sqrt{1-(v/c)^2}} \\ y' = y \\ z' = z \\ t' = \dfrac{t - xv/c^2}{\sqrt{1-(v/c)^2}} \end{cases}$$

定义 $x_4 = \mathrm{i}ct$，构成复四维空间. 令 $\gamma = 1/\sqrt{1-\beta^2}$, $\beta = v/c$

$$\begin{pmatrix} x_1' \\ x_2' \\ x_3' \\ x_4' \end{pmatrix} = \begin{pmatrix} \gamma & 0 & 0 & \mathrm{i}\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\mathrm{i}\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

**速度变换**

$$u_x' = \frac{u_x - v}{1 - \dfrac{v}{c^2} u_x} \qquad u_y' = \frac{u_y}{\gamma\left(1 - \dfrac{v u_x}{c^2}\right)} \qquad u_z' = \frac{u_z}{\gamma\left(1 - \dfrac{v u_x}{c^2}\right)}$$

## 运动尺度的缩短

$$l=\sqrt{1-(v/c)^2}\,l_0=l_0/\gamma$$

## 运动时钟的延缓

$$\Delta t=\frac{\Delta\tau}{\sqrt{1-(v/c)^2}}=\gamma\Delta\tau$$

## 四维速度矢量

$$U_\mu=\frac{\mathrm{d}x_\mu}{\mathrm{d}\tau}=\gamma_u(\boldsymbol{u},\mathrm{i}c)$$

## 四维波矢量

$$k_\mu=\left(\boldsymbol{k},\mathrm{i}\frac{\omega}{c}\right)$$

相位 $\varphi=k_\mu x_\mu=k'_\nu x'_\nu$ 为 Lorentz 不变量.

## 四维波矢量变换

$$\begin{cases}k'_x=\gamma\left(k_x-\dfrac{v}{c^2}\omega\right)\\ k'_y=k_y\\ k'_z=k_z\\ \omega'=\gamma(\omega-vk_x)\end{cases}$$

## 横向 Doppler 效应

取 $\omega'=\omega_0$

$$\omega=\frac{\omega_0}{\gamma\left(1-\dfrac{v}{c}\cos\theta\right)}$$

取 $\theta=\dfrac{\pi}{2}$

$$\omega=\frac{\omega_0}{\gamma}$$

## 四维电流密度矢量

$$J_\mu=\gamma_u(\rho_0\boldsymbol{u},\mathrm{i}c\rho_0)=(\boldsymbol{J},\mathrm{i}c\rho)$$

## 四维势矢量

$$A_\mu = \left(\boldsymbol{A}, \frac{\mathrm{i}}{c}\varphi\right) \qquad (A'_\mu = a_{\mu\nu}A_\nu)$$

## 电磁场张量

$$F_{\mu\nu} = \frac{\partial A_\nu}{\partial x_\mu} - \frac{\partial A_\mu}{\partial x_\nu}$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & B_z & -B_y & -\mathrm{i}E_x/c \\ -B_z & 0 & B_x & -\mathrm{i}E_y/c \\ B_y & -B_x & 0 & -\mathrm{i}E_z/c \\ \mathrm{i}E_x/c & \mathrm{i}E_y/c & \mathrm{i}E_z/c & 0 \end{pmatrix}$$

## 相对论电磁场变换公式

$$E'_x = E_x \quad E'_y = \gamma(E_y - vB_z) \qquad E'_z = \gamma(E_z + vB_y)$$

$$B'_x = B_x \quad B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right) \quad B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

## 四维动量

$$p_\mu = mU_\mu$$

$$p_i = \gamma m v_i \ \ (i = 1, 2, 3) \qquad p_4 = \mathrm{i}c\gamma m = \frac{\mathrm{i}}{c}\frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

## 物体总能量

$$W = \frac{mc^2}{\sqrt{1 - v^2/c^2}}$$

## 动能

$$T = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

## 相对论力学方程

$$K_\mu = \frac{\mathrm{d}p_\mu}{\mathrm{d}\tau} \qquad K_\mu = \left(\boldsymbol{K}, \frac{\mathrm{i}}{c}\boldsymbol{K}\cdot\boldsymbol{v}\right)$$

可分解为

$$\boldsymbol{K} = \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}\tau} \qquad \boldsymbol{K}\cdot\boldsymbol{v} = \frac{\mathrm{d}W}{\mathrm{d}\tau}$$

改为对 $t$ 的微分

$$\sqrt{1-\frac{v^2}{c^2}}\,\boldsymbol{K}=\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}\qquad\sqrt{1-\frac{v^2}{c^2}}\,\boldsymbol{K}\cdot\boldsymbol{v}=\frac{\mathrm{d}W}{\mathrm{d}t}$$

定义力 $\boldsymbol{F}=\sqrt{1-\dfrac{v^2}{c^2}}\,\boldsymbol{K}$, 则相对论力学方程化为

$$\begin{cases}\boldsymbol{F}=\dfrac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}\\ \boldsymbol{F}\cdot\boldsymbol{v}=\dfrac{\mathrm{d}W}{\mathrm{d}t}\end{cases}$$